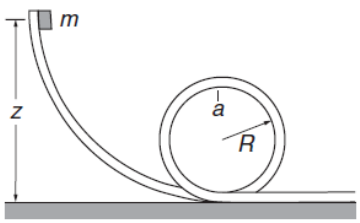
PHY101: Introduction to Physics-1

Monsoon, 2018

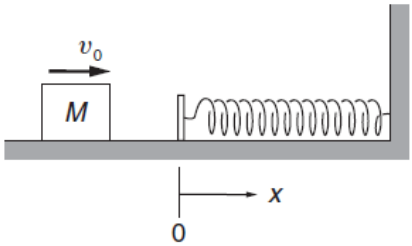
Tutorial-6

1. A block with a mass of 5.7 kg slides on a frictionless table with a speed of 1.2 m/s. It is brought to rest by compressing a spring in its path. By how much is the spring compressed if its force constant is 1500 N/m? (Help: compress 7.4 cm)

2. A small block of mass *m* starts from rest and slides along a frictionless loop-the-loop as shown in the figure. What should be the initial height *z*, so that *m* pushes against the top of the track (at *a*) with a force equal to its weight?
(*Help: z = 3R*)



3. A block of mass *M* slides along a horizontal table with speed v_0. At $x = 0$ it hits a spring with spring constant *k* and begins to experience a friction force, as indicated in the figure. The coefficient of friction is variable and is given by $\mu = bx$, where *b* is a constant. Find the distance *l* the block travels before coming to rest. (*Help:* $l = v_0\sqrt{\frac{M}{Mgb+k}}$)



4. A particle of mass *m* moves in one dimension along the positive *x* axis. It is acted on by a constant force directed toward the origin with magnitude *B*, and an inverse-square law repulsive force with magnitude A/x^2.
(*a*) Find the potential energy function $U(x)$.
(*b*) Sketch the energy diagram for the system when the maximum kinetic energy is $K = \frac{1}{2}m{v_0}^2$
(*c*) Find the equilibrium position, x_0. (*Help:* $total\ U = Bx + \frac{A}{x}, x_0 = \sqrt{\frac{A}{B}}$)